ALGEBRA 2 · RATIONAL FUNCTIONS

Asymptote & Hole Explorer

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Asymptote & Hole Explorer** on Mr. K's Classroom: Algebra 2 → Rational Functions → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. $f(x) = \frac{(x-2)(x+1)}{(x-2)}$. What do you think happens at $x = 2$?

There is a hole: the factor cancels, but x = 2 still isn't allowed.

2 Explore use the tool and record what you see

Build each function

Use the factor switches and sliders.

Function	Holes	Vertical asymptotes	Horizontal asymptote	x-intercepts
$\frac{(x+2)}{(x-2)}$	none	x = 2	y = 1	(–2, 0)
$\frac{(x-2)(x+1)}{(x-2)}$	(2, 3)	none	none	(–1, 0)
$\frac{1}{((x-1)(x+4))}$	none	x = 1, x = –4	y = 0	none
$\frac{3(x+1)}{(x-3)}$	none	x = 3	y = 3	(–1, 0)
$\frac{(x+3)(x-1)}{((x-1)(x-5))}$	(1, –1)	x = 5	y = 1	(–3, 0)

3 Write the rule fill in each blank from what you observed

1. A factor that cancels from the top and bottom makes a **hole**.
2. A factor left in the denominator makes a **vertical asymptote**.
3. Top degree < bottom degree: horizontal asymptote **y = 0**. Equal degrees: y = **the ratio of leading coefficients**. Top degree > bottom degree: **no horizontal asymptote**.
4. The x-intercepts come from the factors left in the **numerator**.

4 Check no tool: show your work

1. Find the hole, asymptotes and x-intercept of $f(x) = \frac{(x+4)(x-2)}{((x-2)(x+1))}$.

Hole (2, 2); VA x = –1; HA y = 1; x-intercept (–4, 0)

2. Find the horizontal asymptote of $\frac{5x^2}{(x^2-9)}$.

y = 5

3. Find the vertical asymptotes of $\frac{2}{(x^2 - 16)}$.

x = 4 and x = −4

5 Reflect

Why is a hole different from a vertical asymptote?

At a hole the canceled factor makes the function undefined at one point only; nearby values approach a finite number. At an asymptote the denominator still goes to 0, so the values blow up.